

Lab 10 - TCP/IP and Three-Way Handshake



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Problem Overview

Containerized Hosts

- client.
- server.
- sniffer.

Tasks

- 1 Access the client host and verify network connectivity with the server using the command: `ping -c 3 server`.
- 2 From the client, clear the ARP cache using the command `ip neigh flush all`. Subsequently, force ARP discovery by executing `ping -c 1 server`. Then, verify the MAC address associated with the server by running `ip neigh show`.
- 3 Access the server and start an HTTP service using the following command: `python3 -m http.server 80`. In a separate terminal session, access the client host and issue the request: `curl -v http://server`. Record and analyze the response returned by the web service.
- 4 On the sniffer host, use `tcpdump` to monitor TCP traffic between the client and the server by executing: `tcpdump -i any 'tcp and host and host' -n -vv`. In another terminal session on the server, execute: `nc -l -p 8080`. Then, from the client, establish a connection using `nc server 8080` and transmit a text message. Analyze the captured packets and describe the observed TCP three-way handshake process.
- 5 After completing the previous step, execute the following command on the client: `nc -vz server 9090`. A connection failure is expected. Examine the packet capture in `tcpdump` during this process and describe the observed TCP behavior.
- 6 From the client, perform a port scan against the server using the following command: `nmap -Pn -p 80,8080 server`. Evaluate the open ports and associated services based on the scan results.

Tools

- The following tools are available for completing OTLab 10: `curl`, `ifconfig`, `ip`, `ipcalc`, `nc`, `nmap`, `ping`, `python3`, `tcpdump`, and `traceroute`.

Nomenclature

- ARP: Address resolution protocol.
- HTTP: Hypertext transfer protocol.
- IP: Internet protocol.
- MAC: Media access control.
- TCP: Transmission control protocol.